

7. Technical Decisions

This document records important design decisions.

Algorithm Choice

Decision:

Use **A*** algorithm for route planning.

Reason:

Provides efficient pathfinding with moderate implementation complexity.

Map Data Source

Decision:

Use OpenStreetMap for map tiles and geographic data.

Reason:

OpenStreetMap provides free global geographic data suitable for development and testing.

Obstacle Source

Decision:

Use Overpass API to retrieve building geometries.

Reason:

Buildings represent realistic physical obstacles for drones.

Visualization

Decision:

Use a 2D map interface instead of full 3D visualization.

Reason:

Reduces system complexity while maintaining functional route planning.

Height Representation

Decision:

Represent altitude as a numeric parameter in the system.

Reason:

Allows algorithmic handling of height without requiring 3D rendering.

Communication

Decision:

Use REST API instead of WebSockets.

Reason:

Mission commands and route generation do not require continuous streaming.